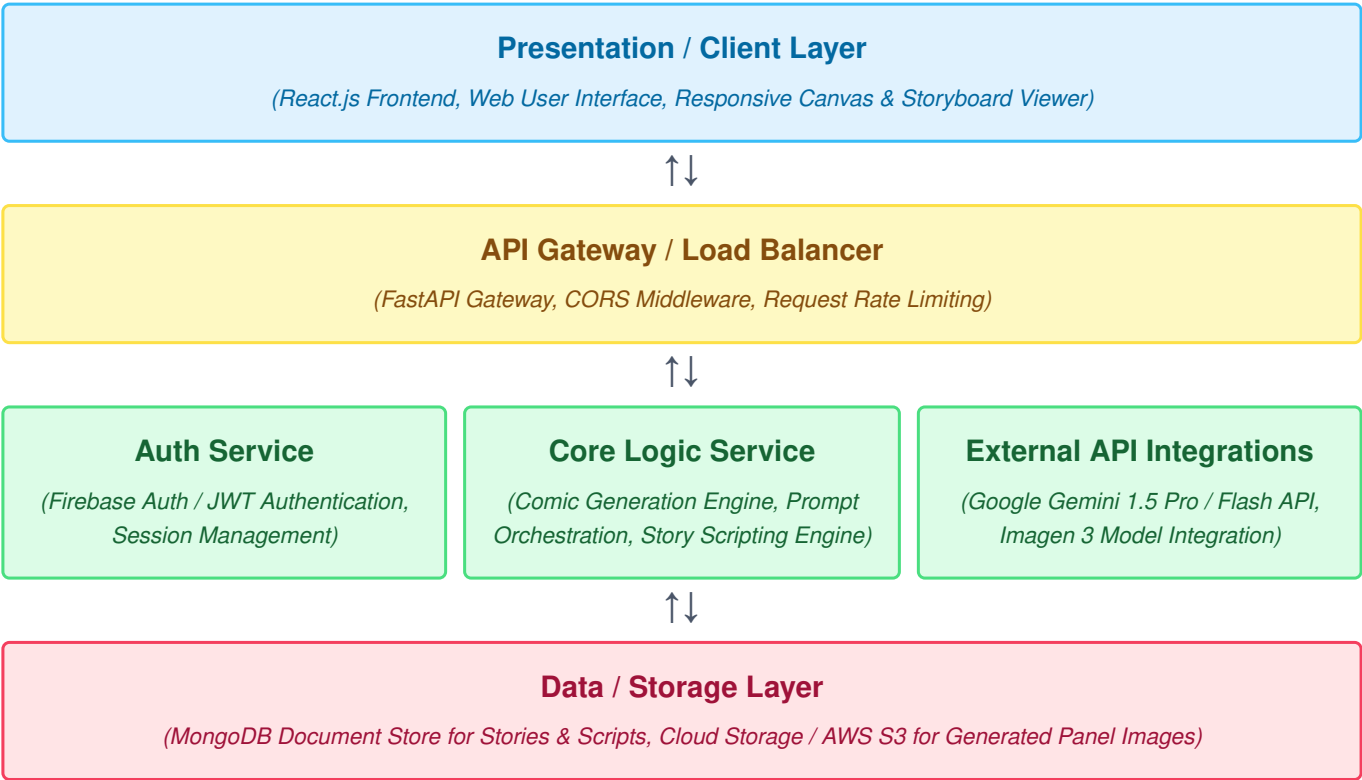


SOLUTION ARCHITECTURE

Date	29-09-2026
Team ID	SWTID-2026-6684
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

COMPONENT NAME	DESCRIPTION / ROLE IN ARCHITECTURE	TECHNOLOGIES USED
Presentation Layer	Provides an interactive user interface allowing creators to input story prompts, select comic art styles (e.g., Manga, Graphic Novel, Cartoon), customize panel layouts, edit generated dialogue bubbles, and download/export finished PDF or digital comic books.	React.js, Tailwind CSS, HTML5 Canvas, Axios
API Gateway	Acts as the single entry point for client requests. Handles route management, user session token verification, request validation, rate limiting, and forwards incoming traffic efficiently to internal microservices.	FastAPI (Python), CORS Middleware, Uvicorn Server
Microservices (Core Logic & Auth)	• Auth Service: Manages secure user signup, login, session validation, and user profile metadata. • Core Logic Service: Handles story breakdown into structured comic panels, dialogue generation, image prompt enrichment with character consistency cues, and PDF comic compilation.	Python, Firebase Auth / PyJWT, LangChain, Pillow (PIL), ReportLab
External API Integrations	Integrates with Google's advanced AI models: • Google Gemini Models: Generates multi-panel plotlines, scene scripts, character descriptions, and dynamic dialogues. • Imagen 3 / Image Gen APIs: Converts detailed panel text descriptions into high-resolution comic visual panels.	Google GenAI SDK, Gemini 1.5 Flash/Pro API, Imagen 3 API
Database & Cloud Storage	Stores structured document records for user profiles, comic projects, scene metadata, and script text. Uses cloud blob storage to securely store, serve, and cache generated image panels and rendered comic PDFs.	MongoDB Atlas, AWS S3 / Firebase Cloud Storage